

# CISC 3667: Game Design and Development

## Group Project 1: Race to the End!

**Assigned:** Thursday, September 17

**Playtest:** Thursday, September 24

**Due:** Tuesday, September 29 **before class**

**Groups:** 5 students each (one group will have 6 students due to class size)

## Project Overview

In this project, your group will **design, prototype, playtest, and revise a simple race-to-the-end board game.**

This project primarily focuses on **game rules and systems**. You will practice designing rules that establish the game's space, player actions, choices, interactions, consequences, and conditions for winning.

As you design your game, you will consider three types of rules:

- **Constitutive / Foundational Rules:**
  - The game's underlying formal structure: the relationships, states, values, and systems that determine how the game works and changes.
- **Operational / Written Rules:**
  - The explicit rules that tell players what they can and cannot do and how to play.
- **Implicit / Behavioral Rules:**
  - The unwritten expectations that shape player behavior, such as sportsmanship, etiquette, and appropriate conduct.

You will then **playtest your game with another group**, identify problems or opportunities in your design, and revise your game based on what you observe.

The goal is not to create the most complicated game possible. The goal is to create a **clear, playable, and engaging game system in which the rules work together to produce an intended player experience.**

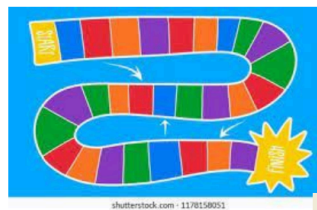
# Project Schedule

|             |                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------|
| Sept. 17    | <b>Project Assigned:</b> Form groups and begin designing your game.                                              |
| Sept. 24    | <b>Playtest:</b> Bring a playable prototype to class and playtest it with another group.                         |
| Sept. 25–28 | <b>Revise &amp; Complete:</b> Use playtest feedback to revise your game and complete your report and reflection. |
| Sept. 29    | <b>Final Submission:</b> Submit the Group Game Design Report and Individual Reflection to Brightspace.           |

## Part 1: Design Your Game

### Step 1: Design the Game Space

Create a board or path with a clearly identified **start and end point**.



Your path can take many forms, including:

- Linear
- Branching
- Looping
- Multiple paths
- A combination of different path types

Clearly identify the important spaces on your board.

These might include:

- Start
- Finish
- Special spaces
- Obstacles
- Shortcuts
- Alternate paths
- Reward spaces
- Penalty spaces
- Spaces where players interact

Think about your board as part of the game's rules. The board's structure determines where players can go and what choices they have.

### **Consider:**

- Where do players start?
- Where are they trying to go?
- Are there multiple ways to reach the end?
- Are some paths safer than others?
- Are some paths faster but riskier?
- Can players affect one another?
- Does the board create meaningful decisions?

## **Step 2: Goal, Theme, and Narrative**

Decide what players are trying to accomplish and give your game a theme or context.

Your objective should answer:

**What are players trying to accomplish?**

Your winning condition should answer:

**How does a player know they have won?**

The theme does not need to be complicated. It should provide context for the game's rules and player experience.

## Step 3: Game Rules

Design the rules that determine how your game works and how players interact with it.

### Constitutive / Foundational Rules

Identify the underlying systems, relationships, states, and values that make your game work.

Consider:

- What important game states exist?
- What values can change?
- How do they change?
- What causes those changes?
- How is movement determined?
- How are outcomes determined?
- What conditions lead to victory?

You do not need to create a formal mathematical model.

### Operational / Written Rules

Write the explicit rules players need in order to play.

Your rules should clearly explain:

- Setup
- Turn order
- What happens during a turn
- Player actions
- Movement
- Restrictions
- Consequences
- End and winning conditions

Avoid vague instructions. Players should be able to determine what happens without needing the designers to explain every situation.

Consider unusual situations, such as:

- Two players occupying the same space
- A player being unable to move

- A player running out of a resource
- Two players reaching the winning condition at the same time
- An unexpected result from a game component

## Implicit / Behavioral Rules

Consider the behaviors you expect players to follow even though they are not explicitly written in the rules.

These might include:

- Good sportsmanship
- Fair play
- Appropriate turn-taking
- Allowing other players to participate
- Not intentionally distracting another player
- Not deliberately damaging game components
- Not intentionally delaying the game

Ask:

**Is there an implicit expectation that should actually become an explicit written rule?**

## Step 4: Player Actions and Choices

Identify the important things players can **do** in your game.

Examples include:

- |          |                    |
|----------|--------------------|
| • Move   | • Take             |
| • Roll   | • Give             |
| • Draw   | • Challenge        |
| • Choose | • Help             |
| • Trade  | • Use an item      |
| • Block  | • Spend a resource |

For important actions, consider:

- |                               |                             |
|-------------------------------|-----------------------------|
| • Who can perform the action? | • What restrictions apply?  |
| • When can they perform it?   | • What happens as a result? |

Also consider whether players have **meaningful choices**.

For example:

**Safe choice:** Move 2 spaces automatically.

**Risky choice:** Roll a die. On 1–3, move 1 space; on 4–6, move 5 spaces.

Ask:

**Why would a player choose one option instead of another?**

If one option is always better, the choice may not be meaningful.

## Step 5: Player Interaction

Decide how players can interact with one another, if appropriate.

Interaction could include:

- Helping another player
- Hindering another player
- Blocking another player
- Speeding up or slowing down another player
- Swapping positions
- Competing for limited resources
- Choosing whether to help or hinder another player

Your game does not need to include lots of interaction. Any interaction should support the game's goal and intended experience.

## Step 6: Chance, Skill, Risk, and Reward

Consider the role of **chance and skill** in your game.

If your game uses dice, cards, spinners, or other random elements, consider:

- What is random?
- How much control do players have?
- Does chance create interesting uncertainty?
- Can players make meaningful decisions around uncertainty?

You are **not expected to calculate probabilities** for this project.

Also consider **risk and reward**:

- What risks can players take?
- What rewards can they receive?
- Why might a player choose a risky option?

## Step 7: Player Experience

Consider the experience you want players to have.

Ask:

- Is the objective clear?
- Is the game too easy or difficult?
- Can players understand what they are supposed to do?
- Can players recover from setbacks?
- Does the game remain engaging for players who are behind?
- Does the game create interesting decisions?
- Is the challenge appropriate?
- Do players understand why they won or lost?

## Step 8: Prepare Your Prototype

Before **September 24**, create a playable prototype.

Your prototype should include:

- |                                                         |                      |
|---------------------------------------------------------|----------------------|
| • Game board/path                                       | • Setup instructions |
| • Player pieces                                         | • Turn structure     |
| • Cards, dice, tokens, or other<br>necessary components | • Movement rules     |
| • Written rules                                         | • Player actions     |
|                                                         | • Winning condition  |

The prototype does **not** need to be polished. Its purpose is to test your game.

## Part 2: Playtest and Peer Feedback

On **September 24**, your group will exchange games with another group. Each group will play the other group's game **without assistance from the original designers**.

Your game should be complete enough that another group can:

- Set up the game
- Read the instructions
- Understand the rules
- Begin playing
- Complete the game
- Determine who won

**without asking the original designers for clarification.**

The original designers should **not explain, correct, or guide the other group during the playtest**. The group playing your game will provide written feedback about their experience, which you will use to identify areas for improvement.

### What the Playing Group Should Do

After playing another group's game, answer the following questions and return your feedback to the original designers.

#### Rules & Instructions

1. **Were you able to set up the game using only the provided instructions?** If not, what was unclear?
2. **Were the rules clear enough to begin playing without help from the designers?**
3. **Did you encounter any rules that were confusing, ambiguous, incomplete, or contradictory?** Identify them.
4. **Were there any situations during play that the rules did not explain?** If so, describe what happened.

#### Gameplay & Choices

5. **Were the available player actions clear?** Did you understand what you could and could not do?
6. **Did the game give you meaningful choices?** Describe one choice you made and why you made it.



7. **Did you discover any unexpected strategies or ways of playing?**
8. **Did the game behave as you expected based on the rules?** If not, describe what happened.

### **Player Experience**

9. **What did you enjoy about the game?**
10. **Was anything confusing, frustrating, boring, or particularly engaging?**
11. **Did the game feel too short, too long, or about the right length?**
12. **Was it clear when and why someone won?**

### **Overall Feedback**

13. **What is the strongest part of the game?**
14. **What is the most important thing the designers should change or improve?**

### **Important**

The purpose of this playtest is to determine whether the game can **stand on its own**. If players need the designers to explain a rule in order to continue playing, that is valuable feedback about the game's instructions or design.

### **Revise**

After the playtest, make **at least one meaningful design change**.

Your revision could involve:

- A constitutive/foundational rule
- An operational/written rule
- An implicit/behavioral rule
- Movement
- The board layout
- Player interaction
- The winning condition
- A game component
- Chance or randomness
- Risk and reward
- A confusing rule

Your report will explain **what you changed, what you observed, and why you made the change**.

## Part 3: Group Game Design Report (70 points - One submission per group)

Submit a **4-6 page report or equivalent number of slides** to Brightspace.

Your report should document your game and analyze the design decisions your group made.

### 1. The Game

Include:

- **Game Title**
- **Number of Players**
- **Objective**
- **Theme/Narrative**
- **Board/Path** — include a photo or digital image
- **Components**
- **Setup**
- **Turn Order**
- **Movement & Turn Rules**
- **Player Interaction**
- **End/Win Condition**

### 2. The Three Types of Rules

#### Constitutive / Foundational Rules

Explain the **constitutive/foundational rules that underlie your game**.

For each relevant rule or relationship, explain:

- What system, state, value, or relationship it describes
- How it changes
- What causes the change
- How it affects gameplay

#### Operational / Written Rules

Explain the **operational/written rules of your game**.

Explain:

- What players are required or allowed to do
- When each rule applies
- What happens as a result
- Why the rule is necessary

Include any **written rules that you revised after playtesting** and explain why you changed them.

### **Implicit / Behavioral Rules**

Explain the **implicit/behavioral rules or expectations in your game**.

Identify:

- What behavior you expect from players
- Why it is important
- Whether it should remain implicit or become an explicit written rule

### **3. Actions, Choices, and Consequences**

Identify at least **two important player actions** and explain their restrictions, consequences, and strategic possibilities.

Then identify at least **one meaningful choice** and explain:

- What the player chooses between
- Why both options are potentially useful
- What consequences result
- How the choice affects the game

### **4. Chance, Skill, Risk, Reward, and Player Experience**

Explain how your game uses **chance, skill, risk, and reward**, where applicable.

Discuss:

- Player control
- The role of randomness
- Meaningful decisions around uncertainty
- Risks and rewards
- Effects of setbacks
- The intended player experience

You are **not expected to calculate probabilities or formally analyze game balance** for this project.

## 5. Playtesting and Iteration

Describe your playtest and revision.

Include:

- **What you observed:** What happened when another group played your game?
- **What you learned:** What did the playtest reveal about your rules or player experience?
- **What you changed:** What meaningful change did you make?
- **Why you changed it:** What observation led to the change?
- **Result:** How did the revision affect the game?

## 6. Theoretical Analysis

### A. What Is a Game?

Choose at least **two** of the four definitions of a game discussed in class:

- Kevin Maroney
- Jane McGonigal
- Jesse Schell
- Chris Crawford

For each, briefly explain the relevant idea and apply it to your game.

### B. Types of Fun

Identify at least **two Types of Fun** from LeBlanc's framework.

For each, explain:

- What Type of Fun it is
- What part of your game creates that experience
- Why players might enjoy it

### C. Bartle Player Types

Identify at least **one Bartle Player Type** who might enjoy your game.

Explain:

- Which Player Type you selected
- What motivates that type
- What features of your game would appeal to them

## Group Game Design Report Rubric (70 Points)

| Category                                                   | Criteria                                                                                                                                                          | Points    |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Board/Path Design</b>                                   | Clear start and end; understandable path; functional board that supports the game's rules and player experience.                                                  | <b>7</b>  |
| <b>Theme &amp; Objective</b>                               | The objective is clear; the theme is coherent and supports the intended experience.                                                                               | <b>5</b>  |
| <b>Rules &amp; Game System</b>                             | Constitutive/foundational, operational/written, and implicit/behavioral rules are clearly identified and explained. Rules are workable, consistent, and playable. | <b>12</b> |
| <b>Player Actions, Choices &amp; Interaction</b>           | Player actions are clearly defined; meaningful choices are present; player interaction supports the design.                                                       | <b>8</b>  |
| <b>Chance, Skill, Risk, Reward &amp; Player Experience</b> | Chance, skill, risk/reward, and their effects on the player experience are thoughtfully considered where appropriate.                                             | <b>6</b>  |
| <b>Win Condition &amp; Playability</b>                     | End condition is clear; players can determine who wins; the game can be played using the provided rules.                                                          | <b>3</b>  |
| <b>Playtesting &amp; Iteration</b>                         | Playtest is documented; problems or opportunities are identified; at least one meaningful revision is made based on observations.                                 | <b>7</b>  |
| <b>Definitions of a Game</b>                               | Applies at least two course definitions accurately and thoughtfully.                                                                                              | <b>7</b>  |
| <b>Types of Fun</b>                                        | Identifies at least two Types of Fun and connects them to the game.                                                                                               | <b>5</b>  |
| <b>Bartle Player Type</b>                                  | Identifies at least one appropriate Player Type and explains why that type would enjoy the game.                                                                  | <b>4</b>  |
| <b>Completeness, Clarity, &amp; Organization</b>           | Report is approximately 4-6 pages or equivalent slides; writing is clear and organized; all required sections are included.                                       | <b>6</b>  |
| <b>Total</b>                                               |                                                                                                                                                                   | <b>70</b> |

## **Part 4: Individual Reflection (30 points - One submission per student)**

Submit your own reflection to Brightspace.

### **1. Personal Contribution**

What did you personally contribute to the project? Describe specific tasks, ideas, design decisions, or work that you completed.

### **2. Collaboration**

How did your group work together?

Include:

- One example of something your group did well
- How your group handled a disagreement, different ideas, or an important design decision

### **3. Game Design Reflection**

- What worked well in your game?
- What did not work as well as you hoped?
- What did you learn about your game's rules?
- Did playtesting reveal anything unexpected?
- If you continued developing the game, what would you change?

### **4. Theory Reflection**

- Which Types of Fun did you personally experience while playing the game?
- Which Bartle Player Type do you think best matches the game?
- Why?

### **5. Learning Reflection**

What did you learn about:

- Game design
- Designing rules
- Playtesting and iteration
- Working as a member of a design team

## Individual Reflection Rubric (30 Points)

| Category                      | Criteria                                                                                                  | Points    |
|-------------------------------|-----------------------------------------------------------------------------------------------------------|-----------|
| <b>Personal Contribution</b>  | Clearly describes the student's individual contributions to the project.                                  | <b>10</b> |
| <b>Collaboration</b>          | Provides a meaningful example of collaboration and reflects on group decision-making.                     | <b>5</b>  |
| <b>Game Design Reflection</b> | Reflects thoughtfully on the game's strengths, weaknesses, rules, playtesting, and possible improvements. | <b>5</b>  |
| <b>Theory Reflection</b>      | Reflects on Types of Fun experienced and connects the game to Bartle Player Types.                        | <b>5</b>  |
| <b>Learning Reflection</b>    | Demonstrates meaningful learning about game design, rules, playtesting, iteration, and collaboration.     | <b>5</b>  |
| <b>Total</b>                  |                                                                                                           | <b>30</b> |